

# **Paper D-08: Ecology and Trophic Systems: A Constraint-Driven Flux Dynamics (CDFD) Perspective**

Steve Bico Mujjabi, MD  
Independent Researcher  
ORCID: 0009-0001-0556-5516

May 2026

## Abstract

This paper asks whether Constraint-Driven Flux Dynamics (CDFD) and Adaptive Flux Limitation (AFL) can give the named system a sharper audit language. The working variables are driving flux  $\Phi$ , constraint  $C$ , surface responsiveness  $S$ , and structural memory  $M_s$ . The useful question is plain: what can be measured, what would count as overload, and what result would make the mapping fail? In this paper we apply Adaptive Flux Limitation (AFL) to trophic ecology, modelling energy flow as constrained transport governed by the potential  $\Phi = \Phi_{primary} + \Phi_{detrital} + \Phi_{nutrient}$  and constraint  $C = C_{predation} + C_{competition} + C_{resource}$ . The Lotka-Volterra predator-prey system is the AFL limit cycle; trophic cascade is the AFL cascade propagation criterion applied to food webs; keystone species are AFL constraint hubs. Yellowstone wolf reintroduction is the AFL demonstration experiment for apex constraint restoration.

**Status:** Inferred from AFL first principles; quantitative predictions are hypothesis-level.

**Series tag:** Dom-D08

**Greek-symbol convention.** Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here  $\Phi$  denotes driving flux,  $\Psi_s$  denotes the adaptive operating ratio,  $\Omega$  denotes a domain or state space, and  $\Lambda$  denotes the Life Number where used. The coefficients  $\alpha$ ,  $\beta$ , and  $\gamma$  denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. The symbols  $\delta$  and  $\epsilon$  denote perturbation or tolerance scales, while  $\eta$ ,  $\lambda$ ,  $\mu$ ,  $\nu$ ,  $\rho$ ,  $\sigma$ ,  $\tau$ , and  $\phi$  denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

## 1 Series Context

Part I sets the flow-constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

## 2 Introduction

### 2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [1], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux ( $\Phi$ ) against a rigid topological constraint ( $C$ ) can drive system saturation. When  $\Phi/C$  approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

### 2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory ( $M_s$ ) and its relaxation parameter decays ( $\tau_{relax} \rightarrow \infty$ ), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Trophic networks define the energetic architecture of ecosystems. Primary producers (plants, algae, chemolithoautotrophs) capture energy from sunlight or chemical gradients; herbivores consume producers; carnivores consume herbivores; apex predators consume carnivores. Each trophic transfer is governed by the same principle: flux flows from high-potential to low-potential nodes subject to predation and metabolic constraints.

The AFL ecological framework connects to the Part IV Earth series: biodiversity (Paper 07) is the AFL distributed throughput principle applied to species; ecological energy transport (Paper 06) models the thermodynamic constraints on biomass transfer; pollution (Paper 08) is the AFL perturbation of ecological constraint through toxin accumulation.

The stability of trophic networks has enormous practical implications. The 2004 collapse of the Northwest Atlantic cod fishery eliminated a  $\sim 1$  billion kg/year harvest in under a decade, demonstrating that even well-studied ecosystems can undergo rapid AFL constraint failure. AFL provides a predictive framework for identifying trophic systems near  $\Psi_s^* = 1.2$  before collapse occurs.

### 3 Core Hypothesis

*Trophic networks are AFL systems in which energy flux  $\Phi$  from primary producers to apex predators is constrained by predation pressure  $C$ : the 10% trophic efficiency rule is the AFL  $\Psi_s^* \approx 1$  equilibrium; trophic cascades are AFL constraint-removal events; apex predators are AFL constraint hubs whose removal drives  $\Psi_{s,prey} > 1.2$  in coupled prey populations.*

AFL trophic hypothesis: the Lotka-Volterra predator-prey equations are the AFL limit cycle for a two-species system where predator acts as dynamic  $C$  on prey  $\Phi$ . The stability of real food webs around multiple  $\Psi_s^* \approx 1$  equilibria explains the observed resilience of complex ecosystems: species diversity provides redundant AFL pathways that maintain  $\Psi_s^* \approx 1$  even after single-species perturbations. **Status: Hypothesis; Lotka-Volterra AFL mapping is analytic; multi-trophic extension requires calibration from ecosystem energy flow data.**

### 4 Ecological Potential Fields

We define the generalised trophic potential:

$$\Phi = \Phi_{primary} + \Phi_{detrital} + \Phi_{nutrient} \quad (1)$$

where:

- $\Phi_{primary}$  — primary productivity potential (gross primary production in gC/m<sup>2</sup>/year; driven by solar irradiance, water, and nutrient availability; the fundamental energy input to the trophic AFL system);
- $\Phi_{detrital}$  — detrital energy potential (dead organic matter available for decomposers; often exceeds  $\Phi_{primary}$  in energy transferred, as  $\sim 90\%$  of production enters the detrital rather than grazing food web);
- $\Phi_{nutrient}$  — nutrient cycling potential (N, P, Fe recycled by decomposers; determines the regeneration of  $\Phi_{primary}$  capacity; AFL nutrient cycling is the positive feedback maintaining ecosystem productivity).

## 5 Trophic Constraint Axes

$$C = C_{predation} + C_{competition} + C_{resource} \quad (2)$$

Table 1: Trophic constraint axes and AFL roles.

| Constraint axis   | Ecological substrate                        | AFL role                                          |
|-------------------|---------------------------------------------|---------------------------------------------------|
| $C_{predation}$   | Predator density and kill rate              | Primary trophic constraint; regulates prey $\Phi$ |
| $C_{competition}$ | Intraspecific and interspecific competition | Resource partitioning among taxa                  |
| $C_{resource}$    | Nutrient, light, water limitation           | Primary productivity ceiling                      |
| $C_{disease}$     | Pathogen and parasite load                  | Reduces effective $\Phi_{prey}$                   |
| $C_{climate}$     | Temperature, precipitation, seasonality     | External modulation of all AFL parameters         |

### 5.1 Adaptive Surface Dynamics: Memory and Responsiveness

The operating ratio  $\Psi_s^*$  is mediated by adaptive surface components:

$$\Psi_s^* = (\Phi/C) \cdot S \cdot M_s \quad (3)$$

where  $S$  (Surface Responsiveness) and  $M_s$  (Surface Memory) evolve as:

$$\frac{dS}{dt} = \kappa_s(\Psi_s^* - S) \quad (4)$$

$$\frac{dM_s}{dt} = \Phi S - d_m M_s \quad (5)$$

These components represent the homeostatic capacity of the system to regulate its own operating ratio through memory and responsiveness.

## 6 Flux Law

Trophic energy throughput:

$$J_{trophic} = -\frac{1}{C_{predation}} \nabla \Phi_{biomass} \quad (6)$$

where  $\nabla \Phi_{biomass}$  represents the gradient in biomass energy density between trophic levels (from producer-rich to predator-poor ends of the food web), and  $C_{predation}$  is the predation pressure constraining energy transfer between levels.

The trophic transfer equation at level  $k$ :

$$J_k = \varepsilon_k \cdot J_{k-1}, \quad \varepsilon_k = \frac{\Psi_{s,trophic,k} \cdot C_{metabolic}^{-1}}{1}, \quad (7)$$

where  $\varepsilon_k$  is the trophic transfer efficiency (AFL  $\Psi_s^*/C$  at level  $k$ ). AFL derives  $\varepsilon \approx 0.1$  (10% efficiency) from the metabolic constraint  $C_{metabolic}$  of heterotrophic respiration.

## 7 Conservation Dynamics

Trophic biomass evolves as:

$$\frac{\partial \Phi_k}{\partial t} = \nabla \cdot \left( \frac{1}{C_k} \nabla \Phi_k \right) + S_k - J_{k \rightarrow k+1} \quad (8)$$

where  $S_k$  is biomass production at trophic level  $k$  (photosynthesis for  $k = 0$ ; consumption of level  $k - 1$  for  $k > 0$ ) and  $J_{k \rightarrow k+1}$  is the flux consumed by the next trophic level.

## 8 Adaptive Constraint Dynamics

Predation constraint evolves dynamically:

$$\frac{dC_{predation}}{dt} = \alpha_{eco} |J_{trophic}| - \beta_{eco} C_{predation} \quad (9)$$

where:

- $\alpha_{eco}$ : predation constraint growth from prey availability (high prey abundance supports high predator reproduction, raising  $C_{predation}$ );
- $\beta_{eco}$ : predator mortality and emigration rate (natural mortality, disease, human hunting reduce  $C_{predation}$ ).

## 9 Lotka-Volterra as AFL Limit Cycle

The Lotka-Volterra predator-prey equations:

$$\frac{dN}{dt} = rN - \alpha NP \quad (10)$$

$$\frac{dP}{dt} = \beta NP - \delta P \quad (11)$$

AFL mapping:  $\Phi = N$  (prey biomass flux);  $C = P$  (predator as dynamic constraint on prey);  $J = \alpha NP$  (consumption flux);  $S = rN$  (prey reproduction).

AFL restatement:

$$\frac{d\Phi}{dt} = r\Phi - J = S - J \quad (12)$$

$$\frac{dC}{dt} = \beta J - \delta C = \alpha_{eco} |J| - \beta_{eco} C. \quad (13)$$

This is exactly the AFL master ODE system. The Lotka-Volterra limit cycle is the AFL limit cycle:

- Prey rises ( $\Phi \uparrow$ )  $\rightarrow$  predation flux  $J \uparrow$   $\rightarrow$  predator rises ( $C \uparrow$ )  $\rightarrow$  prey falls ( $\Phi \downarrow$ )  $\rightarrow$  predator declines ( $C \downarrow$ )  $\rightarrow$  prey rises again.

AFL equilibrium:  $\Psi_{s*}^* = r/\alpha = \delta/\beta = (\text{prey growth rate})/(\text{predation rate}) = (\text{predator death rate})/(\text{predator growth rate})$ . At this equilibrium,  $\Psi_{s,trophic} \approx 1$ , consistent with the observed near-constant predator-prey biomass ratios at ecosystem scale.

## 9.1 Adaptive Surface Dynamics: Memory and Responsiveness

The operating ratio  $\Psi_s^*$  is mediated by adaptive surface components:

$$\Psi_s^* = (\Phi/C) \cdot S \cdot M_s \quad (14)$$

where  $S$  (Surface Responsiveness) and  $M_s$  (Surface Memory) evolve as:

$$\frac{dS}{dt} = \kappa_s(\Psi_s^* - S) \quad (15)$$

$$\frac{dM_s}{dt} = \Phi S - d_m M_s \quad (16)$$

These components represent the homeostatic capacity of the system to regulate its own operating ratio through memory and responsiveness.

## 10 Flux–Constraint Number and Trophic States

$$\Psi_{s,trophic} = \frac{J_{consumption}}{C_{predation} + C_{competition}} = \frac{\alpha NP}{C_{total}} \quad (17)$$

### 10.1 Stable Trophic Regime ( $\Psi_s^* < 0.8$ )

Prey biomass well below predator constraint capacity. This regime occurs after large predator populations crash (disease, hunting): temporarily unconstrained prey that haven't yet fully recovered their food chain. AFL predicts rapid prey population growth in this regime, followed by predator recovery ( $C$  buildup).

### 10.2 Optimal Trophic Regime ( $\Psi_s^* \approx 1$ )

Consumption matches productive capacity. This is the ecological limit cycle: prey and predator oscillate around their AFL equilibrium. AFL predicts maximal biodiversity (both prey and predator species are maintained) and maximal ecosystem productivity in this regime.

### 10.3 Trophic Overload Regime ( $\Psi_s^* > 1.2$ )

Consumption exceeds regenerative capacity. Overgrazing, overfishing, or pathogen outbreak drives prey biomass below the minimum viable population. AFL cascade: prey decline  $\rightarrow$  predator starvation  $\rightarrow$  further prey decline (positive feedback). Cod collapse (1990s): fishing pressure  $J_{harvest}$  exceeded  $C_{regeneration}$  for >15 years, driving  $\Psi_{s,cod} > 1.2$  until population collapse.

## 11 The 10% Trophic Efficiency Rule as AFL

The empirical observation that only  $\sim 10\%$  of energy is transferred between trophic levels emerges from AFL metabolic constraint:

$$\varepsilon_{trophic} = \frac{J_{production}}{J_{consumption}} = \frac{\beta_{growth}}{\alpha_{consumption} + \beta_{respiration}}. \quad (18)$$

At AFL  $\Psi_s^* = 1$  (ecological equilibrium):

$$\varepsilon^* = \frac{\Psi_s^* \cdot \beta_{eco}}{\alpha_{eco} + \beta_{eco}} \approx \frac{1 \cdot 0.1}{0.9 + 0.1} = 0.1. \quad (19)$$

AFL derives the 10% rule from the ratio of production to respiration at the AFL equilibrium, where  $\alpha_{eco}/\beta_{eco} \approx 9$  (9 units of consumed energy lost to respiration per unit incorporated into predator biomass). This is the AFL prediction for the Lindeman efficiency.

## 12 Trophic Cascade Theorem

*AFL Trophic Cascade Theorem: removal of apex constraint  $C_{apex}$  drives  $\Psi_{s,prey} > 1.2$ , triggering bottom-up cascade collapse that propagates to producers via overgrazing.*

Formal statement:

$$C_{apex} \rightarrow 0 \Rightarrow \Psi_{s,prey} = \frac{J_{prey,reproduction}}{C_{apex,remnant}} \gg 1 \Rightarrow N_{prey} \gg K_{prey} \Rightarrow \Phi_{producers} \rightarrow 0. \quad (20)$$

Cascade propagation:

$$\Psi_{s,k} > 1.2 \Rightarrow \Psi_{s,k-1} > 1.2 \quad (\text{bottom-up cascade}), \quad (21)$$

where the cascading  $\Psi_s^*$  excursion propagates downward through the food web.

Yellowstone experiment: grey wolf extirpation (1926) removed  $C_{apex,wolves}$  from Yellowstone. Elk ( $N_{prey}$ ) rose  $\Psi_{s,elk} \gg 1$ ; overgrazing eliminated willow and aspen ( $\Phi_{producers} \rightarrow 0$ ); stream bank erosion followed (loss of riparian constraint); beaver disappeared (no dam material); fish declined (no habitat). Wolf reintroduction (1995) restored  $C_{apex,wolves}$ , initiating a trophic cascade reversal that transformed vegetation, hydrology, and fish populations — the AFL constraint reset demonstration.

## 13 Keystone Species as AFL Constraint Hubs

Keystone species are AFL constraint hubs: species whose  $C_{keystone}$  is disproportionately large relative to their biomass. Removing a keystone species collapses  $C_{total}$  far more than their biomass fraction would predict:

$$\frac{\Delta C_{total}}{\Delta \Phi_{keystone}} \gg \frac{\bar{C}_{total}}{\bar{\Phi}_{community}}. \quad (22)$$

Examples:

- **Sea otters:** maintain  $C_{sea-urchin}$ ; urchin overgrazing destroys kelp forests ( $\Phi_{producers}$ ) when otters are removed;
- **Elephants:** maintain  $C_{tree,savanna}$ ; elephant exclusion drives bush encroachment, reducing grass-dependent herbivore diversity;
- **Mycorrhizal fungi:**  $C_{nutrient-cycling}$  for forest ecosystems; loss from soil acidification drives  $\Phi_{productivity}$  collapse in sensitive forest types.

AFL predicts keystone species from the structural position in the constraint network, not just their biomass: species that occupy rare niche positions (unique  $C$  contributors) are keystones regardless of abundance.

## 14 Spatial Self-Organisation of Ecosystems

Ecosystems organise into spatial AFL patterns:

- **Ecotones:** boundary zones between ecosystem types where  $\nabla\Phi$  is steepest; high species diversity because multiple AFL niches coexist;
- **Productivity gradients:** tropical ecosystems have high  $\Phi_{primary}$  (high solar irradiance, rainfall) and high  $C_{competition}$ ; polar ecosystems have low  $\Phi$  and low  $C$ ; both operate at  $\Psi_s^* \approx 1$  but different throughput levels;
- **Patchiness:** heterogeneity in  $C_{resource}$  (nutrient patches, water availability) creates spatial  $\Psi_s^*$  heterogeneity that maintains diversity via AFL niche partitioning.

## 15 Predictions

1. **Trophic efficiency universality:** AFL predicts 10% efficiency rule from  $\alpha_{eco}/\beta_{eco} \approx 9$  at all trophic levels; systematic deviations should correlate with metabolic rate at each level (endotherms have higher  $\alpha$ , lower  $\varepsilon$ );
2. **Keystone identification:** species with high  $C$ -centrality (constraint hub score) in the trophic network should show disproportionate ecosystem effects when removed; AFL predicts keystone identification from network topology alone;
3. **Collapse threshold:** trophic level  $k$  collapses when  $\Psi_{s,k} > 1.2$  for  $> 3$  consecutive reproductive cycles; AFL predicts collapse timeline from  $\alpha_k$ ,  $\beta_k$ , and  $\Psi_{s,k}$  trajectory;
4. **Recovery timescale:** trophic recovery after constraint removal follows  $\tau = 1/\beta_{eco}$  where  $\beta_{eco}$  is the predator reproduction rate; large mammals have  $\tau \approx$  decades (consistent with Yellowstone wolf recovery trajectory);
5. **Biodiversity-stability AFL:** ecosystem stability (variance of  $\Psi_{s,trophic}$ ) decreases with species diversity because distributed AFL pathways provide redundant constraint channels; AFL predicts stability  $\propto -\ln(\text{Var}(\Psi_s^*))$ , testable from long-term ecological time series.

**Status:** Predictions 1, 3 supported by ecological literature; predictions 2, 4, 5 are AFL-derived hypotheses.

## 16 Computational Interpretation

The companion script `supplementary_dom_08.py` implements AFL trophic dynamics:

- Lotka-Volterra AFL: 2-species limit cycle; AFL equilibrium  $\Psi_s^* = 1$  verified;
- 5-trophic-level system: energy flow from producers to apex; 10% efficiency rule emerged;
- trophic cascade: apex predator removed at  $t = 100$ ; cascade propagation to producers;
- recovery: predator restored;  $\Psi_s^*(t)$  across all levels; recovery timescale  $\tau$ ;
- outputs: limit cycle trajectory, trophic cascade map, biomass time series, efficiency table.



## 17 Discussion

AFL provides a unified framework for trophic ecology: the Lotka-Volterra limit cycle is AFL limit cycle dynamics; the 10% trophic efficiency rule is AFL  $\Psi_s^{**} = 1$  at the metabolic constraint; trophic cascades are AFL cascade propagation; keystone species are AFL constraint hubs; ecosystem stability is distributed AFL throughput.

The AFL ecological framework connects to the entire Part IV Earth series: ecological energy transport (Paper 06) provides the thermodynamic foundation; biodiversity (Paper 07) describes the multi-species AFL distributed throughput principle; ecological collapse (Paper 11) models the full AFL phase transition from functional ecosystem to degraded state.

AFL also connects ecology to socioeconomic systems: fisheries management is AFL constraint optimisation ( $J_{harvest} < \alpha_{eco} \cdot \Phi_{biomass}$ ); ecosystem services are AFL flux commodities; biodiversity loss is AFL constraint erosion with systemic consequences for human food security.

## 18 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions.  $\Phi$  may be a rate of material, energy, information, capital, or signal flow.  $C$  is the bottleneck that slows or redirects that flow.  $S$  measures the system's ability to reconfigure under stress, and  $M_s$  records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

### 18.1 Current Runtime Diagnostic: Universal Network Cascade

The diagnostic script `Part_E.Synthesis/supplementary/run_partiv_discovery.py` keeps this paper tied to the current CDFD Runtime [4], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the May 2026 run, the localized hub-drive stress test ended with hub  $\Psi_s = 14.8$ , peak network  $\Psi_s = 14.8$ , 21 nodes above  $\Psi_s > 1.5$ , and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

## 19 Conclusion

Trophic networks are AFL flux-constraint systems. Energy flows from primary producers to apex predators through successive trophic levels governed by the universal AFL equation; the 10% efficiency rule, Lotka-Volterra limit cycles, trophic cascades, and keystone species effects all emerge from AFL dynamics with ecological  $(\Phi, C)$  interpretation. AFL connects trophic ecology to all other Part IV papers through the universal flux-constraint framework.

## References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.

- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
- [4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdf execution engine, 2026.